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Operation of a Dual Excitation Multiphase Electrostatic Drive by Amplitude-Modulated ac Voltage

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SUMMARY

Dual Excitation Multiphase Electrostatic Drive (DEMED) is a synchronous motor that is driven by electrostatic force. Because of its light weight, thinness, and large power per weight ratio, it has promise as a small-size, high-power motor in the future. Of the three types of driving method that had been formerly developed for the motor, the single frequency method, which drives the motor by applying a three-phase ac voltage to the electrodes of the motor, is the most useful since it needs the least number of phases of power supply. However, this method has problems with operation at very low speed such as 10 μm/s. In order to solve such problems, this paper proposes a novel add-on operation method. In the new method, called modulation drive, the excitation voltage to each phase of the electrodes is amplitude-modulated by a carrier signal with higher frequency before being applied to the electrodes. In tests regarding the single frequency method, the modulation drive successfully drove a motor with typical power operational amplifiers that were boosted by transformers with ferrite cores, and eliminated the force degradation at low drive speed. Additionally, the force generation of modulation was analyzed, and the results agreed very well with the experimental results. © 2000 Scripta Technica, Electr Eng Jpn, 131(4): 78–84, 2000

Key words: Electrostatic motor; electrostatic actuator; electrostatic force.

1. Introduction

Aiming at development of high-power and lightweight electrostatic motors, the authors have been developing various types of such motors. Among these motors, ac drive dual excitation multiphase electrostatic drive (AC DEMED) has the largest power to weight ratio, namely, 230 W/kg [1]. In addition, a linear motor with a large propulsive force of 310 N has already been developed using this technology [2]. Because of this good power generation performance, the motor is expected to take the place of conventional electromagnetic motors in various mechatronic applications. For operation of the motor, the authors have developed three methods of operation, the single frequency method, dual frequency method, and ac–dc method [1]. The single frequency method is the most convenient of the three, since the motor can be driven by applying a three-phase ac voltage to the electrodes at a velocity in proportion to the frequency of the voltage, while the other methods need two frequencies of three-phase ac or a combination of three-phase ac and dc voltages. However, the single frequency method has two problems when it operates the motor at very low speed. One is the impossibility of using a transformer to generate high-amplitude and low-frequency voltages, and the other is diminution of torque or thrust force. This paper proposes a novel operation method, applies it to a prototype linear motor, and reports the thrust force generation performance.

2. Problems of Single Frequency Methods

AC DEMED consists of a pair of plastic sheets, the slider and stator, as shown in Fig. 1. Each sheet contains many lines of electrodes located at regular intervals. The electrodes are parallel, and the intervals for the slider electrodes and the stator ones are the same. Figure 2 shows a schematic of the dual frequency method, which is the most basic and easy-to-understand method of operation among those that the authors have formerly introduced [1]. The dual frequency method employs two sets of three-phase ac voltages with different frequencies. When the three-phase ac voltage is applied to the electrodes in the slider and stator,

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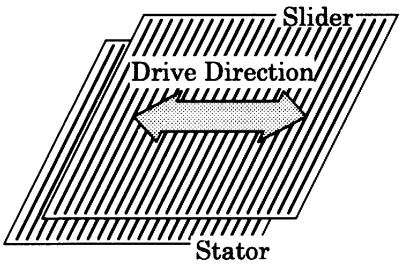


Fig. 1. Overview of DEMED.

the potential distributions illustrated with bold lines in Fig. 2 are induced in the slider and stator, and the distributions generate electrostatic force between them. As a result, the force drives the slider on the surface of the stator and moves it perpendicular to the longitudinal direction of the electrodes. By applying three-phase ac voltages to the electrodes, the two potential distributions run at velocities proportional to the frequencies of the ac voltages. This motor is a kind of synchronous motor, and in the synchronous state, the spatial phase difference between the two distributions is kept constant. Therefore, the synchronous velocity becomes the difference between the velocities at which the two potentials run in the slider and stator.

Currently, most of the experimental prototypes of sliders and stators are fabricated using the same technology as for flexible printed circuit boards, and a typical electrode interval is around 200 μm. The AC DEMED requires that the three-phase voltage have an amplitude around 2 kV to achieve force generation performance comparable to a conventional magnetic motor. Generally, analog amplifiers that can generate such a high voltage are very expensive, and use of the dual frequency method, which requires six such amplifiers, seems impractical.

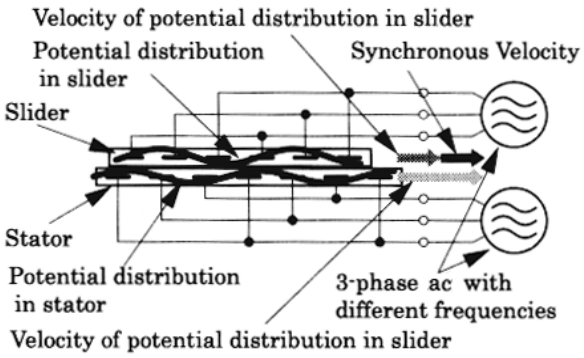


Fig. 2. Schematic cross-sectional view of AC DEMED (dual frequency method).

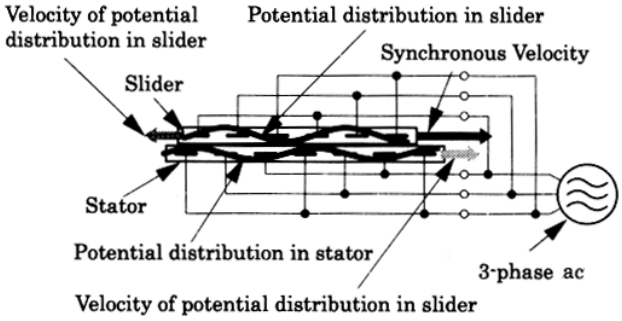


Fig. 3. Single frequency method.

The single frequency method effectively simplifies the driving apparatus for the motor. Figure 3 shows the schematic of the method. To the stator electrodes, three-phase ac voltages are applied. To the slider electrodes, the same three-phase ac voltages are applied, but the order of connection is reversed, as shown in Fig. 2. This connection allows the existence of two potential distributions running in opposite directions at the same speeds in the two elements. As a result, the synchronous running velocity of the slider element is double that at which the distribution in the stator runs. Therefore, the slider runs at a speed proportional to the frequency of the applied ac. The single frequency method reduces the number of high-voltage output amplifiers required for operation of the AC DEMED. In order to reduce the cost for amplifiers, we can use low-voltage amplifiers and transformers as boosters. However, a transformer is not applicable for low slider speed operation with the single frequency method, since such operation decreases the frequency of the applied ac voltage and, as a result, causes saturation of the transformer core. Additionally, in former experiments, a decrease or disappearance of thrust force was observed at low operation speeds such as 10 μm/s. The reason for this phenomenon has not been clarified, but it seems to be the following. Applying a very low frequency to the electrodes of the motor generates a constant-polarity electric field in the insulation between two neighboring electrodes. The field induces a space charge around the electrodes, and the charge neutralizes the charge induced in the electrodes or on their surface and weakens the electric field generated between the slider and stator. This phenomenon becomes an impediment to use of the motor as a servomotor in a system that requires low speed operation, such as a positioning system.

3. Modulation Drive

In order to solve the problems mentioned in the previous section, a novel operation method—modulation

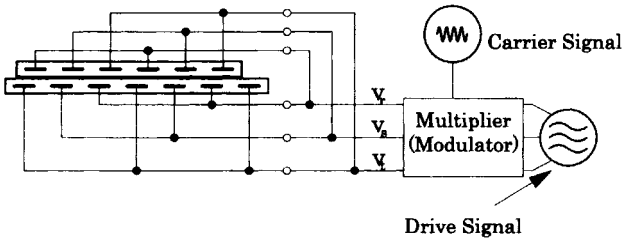


Fig. 4. Application of modulation drive to single frequency method.

drive—has been developed. In this section, the method is introduced and analyzed.

Figure 4 shows a schematic cross-sectional view of the motor and a block diagram of the drive circuit for modulation drive. Modulation drive modulates three-phase ac voltages (drive signal or drive voltage in the following description) with higher frequency (carrier frequency or carrier signal in the following description) and applies them to the electrodes, while the normal single frequency method directly applies the drive voltage to the electrodes. The following formulas show the electrode voltages for both the normal single frequency method and modulation drive.

In the normal single frequency method, the three phases of electrode voltages v_r, v_s, v_t are

$$\begin{cases} v_r = v_0 \sin \omega_s t \\ v_s = v_0 \sin\left(\omega_s t + \frac{2}{3}\pi\right) \\ v_t = v_0 \sin\left(\omega_s t + \frac{4}{3}\pi\right) \end{cases} \tag{1}$$

Here v_0 and ω_s are the amplitude of the excitation voltage and the angular frequency of the drive voltage, respectively. In the case of modulation drive, v_0 is replaced with the carrier voltage, $v_c = v_0 \sin\omega_c t$, as follows:

$$\begin{cases} v_r = v_0 \sin\omega_c t \sin \omega_s t \\ v_s = v_0 \sin\omega_c t \sin\left(\omega_s t + \frac{2}{3}\pi\right) \\ v_t = v_0 \sin\omega_c t \sin\left(\omega_s t + \frac{4}{3}\pi\right) \end{cases} \tag{2}$$

Modulation drive excites the electrodes with voltages that are alternating at the carrier frequency to prevent loss of thrust and saturation of the transformer cores at low drive frequency.

In the following description, we analyze the thrust force of modulation drive and compare it with that of normal drive. To simplify the following calculations, we assume that the potential distributions that are induced by excitation of normal drive in the slider and stator are completely sinusoidal and define them as $v_{sl}(t, x)$ for the slider and $v_{st}(t, x)$ for the stator:

$$\begin{cases} v_{sl} = v_0 \sin\left(\omega_s t + \frac{2\pi}{3p}\left(x - x_{sl}\right)\right) \\ v_{st} = v_0 \sin\left(\omega_s t - \frac{2\pi}{3p}x\right) \end{cases} \tag{3}$$

Here x, x_{sl}, and p are the x position, the slider position, and the interval/pitch at which electrodes are located in the slider and stator, respectively. From these formulas, we obtain the phase difference between the two potential distributions δ as

$$\delta = -2\omega_s t + \frac{2\pi}{3p}x_{sl} + \pi \tag{4}$$

Two distributions between which the phase difference and distance are defined as in Fig. 5 generate an electrostatic force, and its magnitude per unit area over which the electrodes of the slider and stator are facing each other (effective electrode area) is calculated as

$$f_x = \frac{\varepsilon v_0^2}{\lambda^2}\hat{k}_t \sin\delta \tag{5}$$

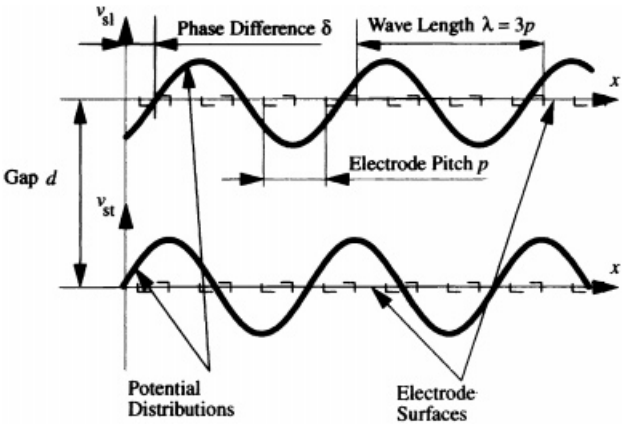


Fig. 5. Phase difference between potential distributions in the slider and stator.

where ε, λ, and $\hat{k}_t$ are the dielectric constant of the space where the motor is situated, the wavelength of the distribution, and a constant that is defined as a function of λ/p [3]. Then, we obtain the thrust force of the motor that has an effective electrode area S as

$$F_x = k_t v_0^2 \sin\left(2\omega_s t - \frac{2\pi}{3p} x_{sl}\right)$$

$$k_t = \hat{k}_t \frac{\varepsilon}{\lambda^2} S \tag{6}$$

The thrust force of modulation drive, $F_{\mathrm{mod}x}$, can be obtained by replacing v_0 in Eq. (6) with $v_0 \sin\omega_c t$ as

$$F_{\mathrm{mod}\,x} = k_t v_0^2 \sin^2\omega_c t \sin\left(2\omega_s t - \frac{2\pi}{3p} x_{sl}\right) \tag{7}$$

Here, we assume that the carrier frequency ω_c is much higher than the drive frequency ω_s. Then we can treat the thrust force as obtainable by averaging $F_{\mathrm{mod}x}$ during a period of the carrier signal. Additionally, we can also approximate the integration used to average $F_{\mathrm{mod}x}$ by integration of just the term in $\sin^2\omega_c t$:

$$\begin{aligned}\overline{F_{\mathrm{mod}\,x}} &= \frac{2\pi}{\omega_c}\int_0^{\frac{2\pi}{\omega_c}} k_t v_0^2 \sin^2\omega_c(t+\tau)\sin\left(2\omega_s(t+\tau) - \frac{2\pi}{3p}x_{sl}\right)d\tau \\ &\approx \frac{2\pi}{\omega_c} k_t v_0^2 \sin\left(2\omega_s t - \frac{2\pi}{3p}x_{sl}\right)\int_0^{\frac{2\pi}{\omega_c}} \sin^2\omega_c(t+\tau)d\tau \\ &\approx \frac{1}{2} k_t v_0^2 \sin\left(2\omega_s t - \frac{2\pi}{3p}x_{sl}\right)\end{aligned} \tag{8}$$

Comparing Eq. (8) with Eq. (7), we find that the thrust force of modulation drive is one-half that of normal drive. This result means that normal drive requires $1/\sqrt{2}$ times the ac amplitude to generate the same thrust force as modulation drive does.

4. Experimental

4.1 Fabrication of tested motor

To test modulation drive, a stacked AC DEMED was fabricated. For the motor, the sliders and stators shown in Fig. 6 were prepared using the same technology as for flexible printed circuit boards. In the slider and stator, electrodes 40 mm long are located at an interval of 200 μm. The effective electrode area is 16 cm^2. To reduce ripple of the thrust force, the electrodes of the slider are skewed [4].

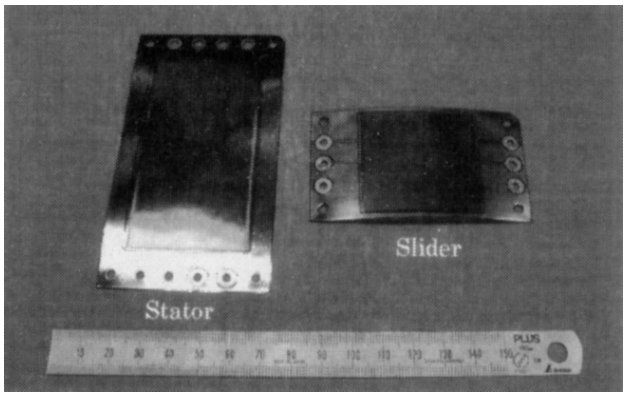


Fig. 6. Stator and slider used in the experiments.

By stacking eight pairs of sliders and stators alternately, a stacked motor was constructed as shown in Fig. 7.

4.2 Operation using transformers

As described before, one of the merits of modulation drive is that amplifiers that cannot generate a high enough voltage for operation of an electrostatic motor can be used successfully by boosting their outputs with transformers. To confirm this advantage, the drive circuit illustrated in Fig. 8 was built, and the motor was operated. The modulated signals were generated by a DSP system and amplified by high-power operational amplifiers (LM12CLK fabricated by National Semiconductor). The amplifier is typically used for the drive of servomotors or high-power speakers and is able to generate a power of 80 W and a voltage of ±25 V when the recommended dc voltage of ±30 V is

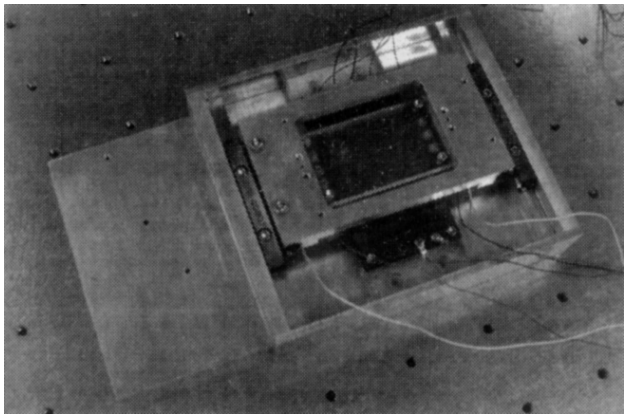


Fig. 7. Stacked motor.

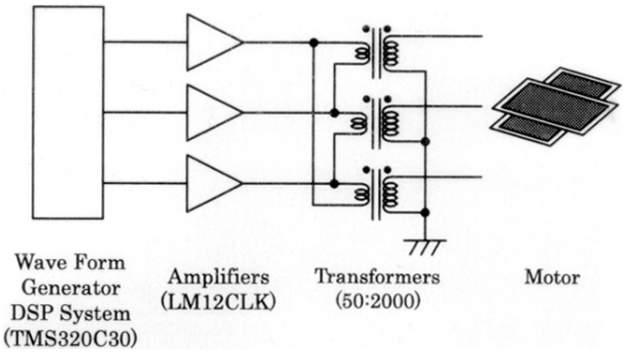


Fig. 8. Driving circuit employing transformers.

supplied. The outputs of the amplifiers were boosted with transformers that have ferrite cores. For the core of each transformer, a ferrite core (PQ50/50 fabricated by TDK) was used. The primary and secondary windings had 50 and 2000 turns, respectively. To construct a three-phase transformer, the legs of the primary and secondary windings of the three transformers were delta connected and star connected, respectively. Finally, the legs of the secondary windings were connected to the electrodes of the motor.

As described before, saturation of the core is a serious problem for generation of high voltages with transformers. Usually, to avoid this problem, three approaches are taken:

1. use higher frequency;
2. increase the number of turns of the windings;
3. insert a gap in the magnetic circuit.

The first approach is the reason why we introduce modulation drive. However, a very high carrier frequency decreases the impedance of the motor, since an electrostatic motor is a capacitive load, and, as a result, the amplifiers

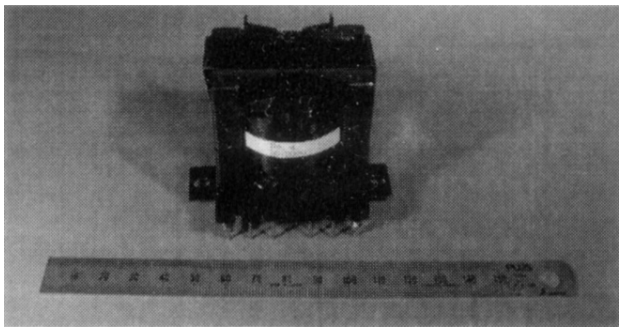


Fig. 9. Transformer employed in the experiments.

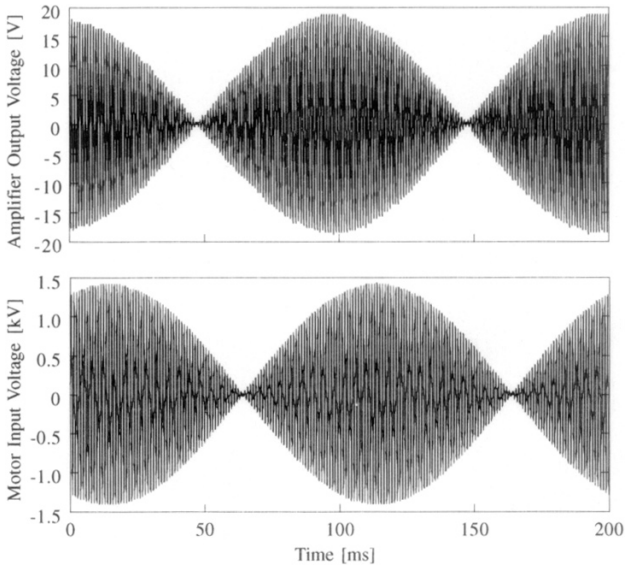


Fig. 10. Voltages of drive output and motor input.

are required to supply a large reactive current. As for turns of the winding, there is a limit because of spacing problems. Therefore, we inserted a gap of 0.6 mm into the magnetic circuit of the transformer yoke to reduce the carrier frequency and size of the transformer.

Figure 10 shows the voltage generated by the amplifier and supplied to the electrodes during a short time span. Figure 11 shows the voltage and current that are generated by the amplifier and supplied to the electrodes. In these measurements, a carrier frequency of 1 kHz was used, and the currents were measured with a current probe (A6302 fabricated by Tektronix).

4.3 Measurement of thrust force

Figure 12 shows the relationship between the thrust force and the amplitude of the applied voltage for modulation drive and normal drive. The thrust force is measured by a simple weight raising experiment. The amplitude is expressed by the effective three-phase voltage. The drive frequency is 1 Hz, which corresponds to a slider synchronous velocity of 1.2 mm/s, and the carrier frequency is 1 kHz. For generation of the electrode voltages, high-voltage amplifiers (model 603C fabricated by TREK Corporation) are employed. Figure 13 shows the relationship between the thrust force and amplitude of the applied voltage for modulation drive and normal drive. To confirm the result of Eq. (8), the voltage amplitude of normal drive is expressed as the effective voltage, that is, $v_0/\sqrt{2}$. As shown in these

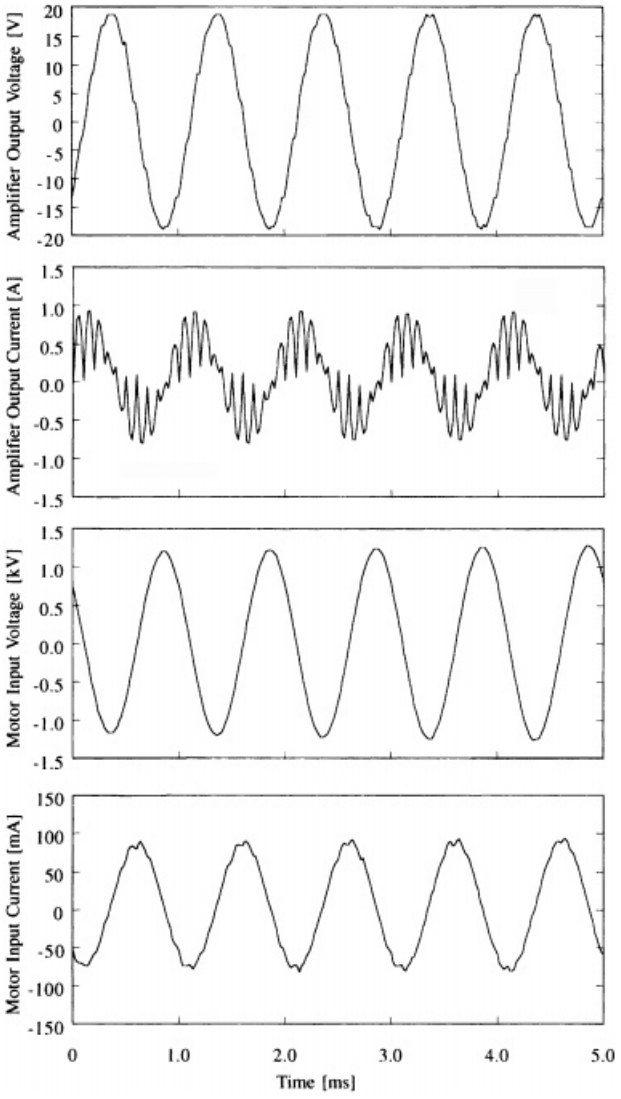


Fig. 11. Voltages and currents of driver output and motor input.

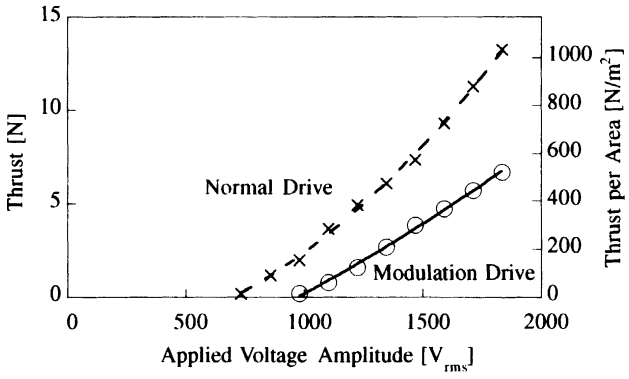


Fig. 12. Relationship between thrust force and applied voltage.

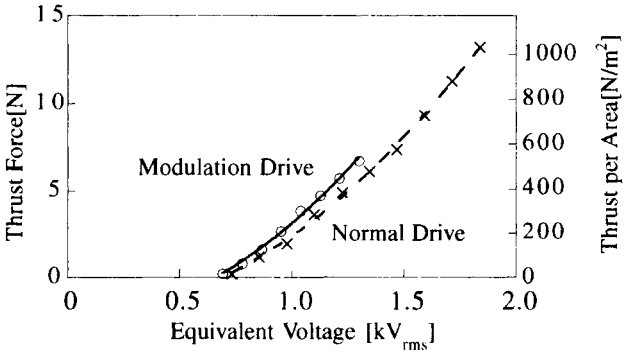


Fig. 13. Relationship between thrust force and equivalent applied voltage.

diagrams, the experimental results agree very well with the analysis expressed by Eq. (8).

4.4 Thrust force at low drive frequency

Figure 14 shows the measured relationship between the thrust force and the drive frequency/slider velocity. In this measurement, a carrier frequency of 1.0 kHz and an applied voltage of 1.5 kV_{rms} are used. While a sudden decrease of thrust force occurred at a low drive frequency of 0.01 Hz in the case of normal drive, the modulation drive system generated a constant thrust force regardless of the drive frequency. Additionally, we did not observe any decrease of thrust force after stopping for 1 hour.

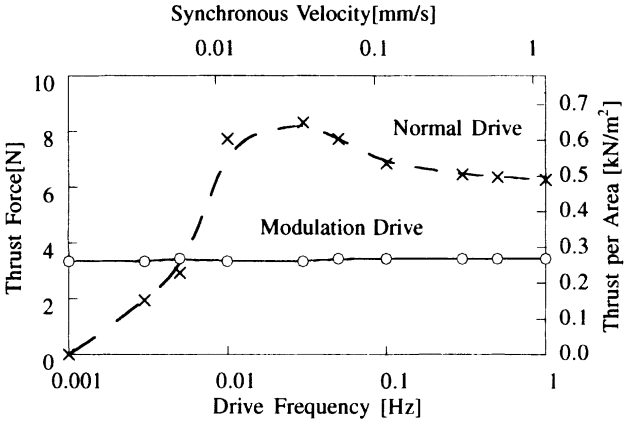


Fig. 14. Thrust force when the motor is driven at very low speed.

5. Conclusions

Modulation drive is introduced as an add-on operation method for ac dual excitation multiphase electrostatic drive. The modulation drive method modulates three-phase ac voltages with much higher carrier frequency and applies them to the electrodes. The thrust force of modulation drive is calculated as the same as that of normal drive when the effective voltage of the latter is considered, and this result agrees very well with experimental results.

Modulation drive successfully prevents the thrust force of the motor from decreasing at very low operation speed. Additionally, the method of operation allows operation of the motor with a combination of transformers and low-voltage-output amplifiers such as audio amplifiers.

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